

NAMMATUTOR GUESS PAPER

Arithmetic Progressions

Board-pattern practice paper, with full solutions

CLASS 10 · MATHEMATICS · CHAPTER 5

QUESTIONS

14

TOTAL
MARKS

34

TIME

70
minutes

PATTERN

CBSE

Arithmetic Progressions — Practice Paper

Class 10 · Mathematics · Chapter 5 · Board pattern

SAMPLE

Max Marks: 34

Time: 70 min

General instructions

1. This paper has five sections — A, B, C, D and E.
2. Section A carries 1 mark per question, Section B 2 marks, Section C 3 marks, Section D 5 marks.
3. Section E is a case-study question carrying 4 marks, with sub-parts.
4. Use $\pi = 22/7$ unless stated otherwise. Show all working for full marks.

SECTION A · 1 MARK EACH

Multiple choice — choose the correct option

1. The common difference of the AP 3, 1, -1, -3, ... is: [1]
(a) 2 (b) -2 (c) 3 (d) -3
2. The 10th term of the AP 2, 7, 12, 17, ... is: [1]
(a) 43 (b) 47 (c) 45 (d) 49
3. If the first term of an AP is 5 and the common difference is 0, all terms of the AP are: [1]
(a) 0 (b) 5 (c) different (d) negative
4. The sum of first 10 natural numbers is: [1]
(a) 45 (b) 50 (c) 55 (d) 60
5. Which term of the AP 21, 18, 15, ... is -81? [1]
(a) 34th (b) 35th (c) 36th (d) 33rd

SECTION B · 2 MARKS EACH

6. Find the 15th term of the AP 10, 7, 4, 1, ... [2]
7. Find the sum of the first 22 terms of the AP 8, 3, -2, ... [2]
8. In an AP, if $a = 5$, $d = 3$ and $a_n = 50$, find n and the sum S_n . [2]

SECTION C · 3 MARKS EACH

9. How many terms of the AP 24, 21, 18, ... must be taken so that their sum is 78? [3]
10. The 4th term of an AP is 0. Prove that its 25th term is three times its 11th term. [3]
11. Find the sum of all three-digit numbers which are divisible by 7. [3]

SECTION D · 5 MARKS EACH

12. The sum of the 4th and 8th terms of an AP is 24, and the sum of the 6th and 10th terms is 44. Find the AP. [5]
13. A sum of Rs 700 is to be used to give seven cash prizes to students of a school for their overall academic performance. If each prize is Rs 20 less than its preceding prize, find the value of each of the prizes. [5]

SECTION E · CASE STUDY · 4 MARKS

14. A hall has 20 rows of seats. The first row has 20 seats, the second row has 22 seats, the third row has 24 seats, and so on — each row having 2 more seats than the one before it. [4]
- This is a classic real-world AP setup — the seat counts per row form an arithmetic progression with $a = 20$, $d = 2$.
- (i) Write the AP formed by the number of seats in each row. [1]
(ii) How many seats are there in the 15th row? [1]
(iii) Find the total number of seats in the hall (all 20 rows). [2]

QUICK REFERENCE – FORMULAS USED IN THIS PAPER

What it finds	Formula
nth term of an AP	$a_n = a + (n - 1)d$
Sum of n terms (last term known)	$S_n = n/2 \times (a + a_n)$
Sum of n terms (last term unknown)	$S_n = n/2 \times [2a + (n - 1)d]$
Common difference	$d = a_n - a_{n-1}$, same for every consecutive pair

Answer Key & Solutions

Class 10 · Arithmetic Progressions

SAMPLE

Step marks shown

SECTION A

1. (b) -2

$$d = a_2 - a_1 = 1 - 3 = -2$$

3. (b) 5

$d = 0$ means every term equals a

5. (b) 35th

$$-81 = 21 + (n-1)(-3) \rightarrow -102 = -3(n-1) \rightarrow n-1 = 34 \rightarrow n = 35$$

2. (b) 47

$$a_{10} = 2 + 9(5) = 47$$

4. (c) 55

$$S_{10} = 10(11)/2 = 55$$

SECTION B

6. $a_{15} = 10 + 14(-3) = -32$ [1 mark: formula, 1 mark: value]

7. $a=8, d=-5, S_{22} = 11 \times (16 + 21 \times (-5)) = 11 \times (-89) = -979$

8. $50 = 5 + (n-1)3 \rightarrow n = 16. S_{16} = 16/2 \times (5+50) = 440$

SECTION C

9. $a=24, d=-3. 78 = n/2 \times (48 + (n-1)(-3)) \rightarrow n^2 - 17n + 52 = 0 \rightarrow n=4$ or $n=13$. Both valid (sum returns to 78 as later negative terms cancel gain) — state **$n = 4$ or $n = 13$** with a note on why both work.

10. $a_4=0 \rightarrow a+3d=0 \rightarrow a=-3d. a_{25}=a+24d=21d. a_{11}=a+10d=7d. 21d = 3 \times 7d \checkmark$ **Proved.**

11. First 3-digit multiple of 7 = 105, last = 994. $n=(994-105)/7+1=128. S=128/2 \times (105+994)=70,336$

SECTION D

12. $a_4+a_8=24 \rightarrow 2a+10d=24. a_6+a_{10}=44 \rightarrow 2a+14d=44$. Subtract: $4d=20 \rightarrow d=5, a=-13$. **AP: -13, -8, -3, 2, ...**

13. $a=\text{first prize}, d=-20, n=7. S_7=700 \rightarrow 7/2 \times (2a-120)=700 \rightarrow 2a-120=200 \rightarrow a=160$. **Prizes: 160, 140, 120, 100, 80, 60, 40**

SECTION E – CASE STUDY

14.

(i) AP: 20, 22, 24, ... with $a = 20, d = 2$.

(ii) $a_{15} = 20 + 14(2) = 48$ **seats**

(iii) $S_{20} = 20/2 \times (2 \times 20 + 19 \times 2) = 10 \times 78 = 780$ **seats**